Abdelrahman Taeha

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EDUCATION

Georgia Institute of Technology — Atlanta, GA
Master of Science in Computer Science – Artificial Intelligence Concentration — *Aug. 2026 – May 2028 (Expected)*

Sacramento State University — Sacramento, CA
Bachelor of Science in Computer Science — *Aug. 2023 – Dec. 2025*

EXPERIENCE

Software Engineer — *May 2025 – Present*
DHCS — *Sacramento, CA*

- Automated monitoring and reporting pipelines across enterprise systems, reducing manual observability overhead by 40% and cutting mean time to detection (MTTD) for production incidents in half through proactive anomaly surfacing.
- Built backend integrations and validation workflows that surfaced system behavior anomalies early, accelerating root-cause analysis by 35% and streamlining cross-team stakeholder communication during incident response.
- Established internal dashboards and reporting applications to visualize agent-driven operational metrics in real time, reducing incident resolution time by 30% and improving leadership visibility into system health.
- Implemented log-processing pipelines and monitoring integrations that improved real-time alerting accuracy by 20%, reducing false positives and engineer alert fatigue.

AI Research Engineer — *Aug. 2025 – Present*
Columbia / Johns Hopkins / Georgia Tech / Stanford Collaboration — *Remote*

- Co-authored **SpineFairBench** (submitted NeurIPS 2026), a counterfactual benchmark evaluating 7,996 paired spinal radiographs to audit demographic sensitivity in VLM-based report generation across a frozen nine-model panel.
- Evaluated medical LLM agent integrity across 840 paired-twin scenarios for the MedInsider FHIR-style benchmark, identifying systematic failure modes in clinical decision-making under simulated institutional pressure.
- Architected evaluation and logging infrastructure — including deterministic action tracking and auditing workflows — for 9 distinct medical vision-language models across full-pipeline and baseline sub-panels.
- Trained a neutral baseline model and developed data collection pipelines for 2 major open-access clinical repositories used to analyze demographic bias in spinal radiology VLM outputs.

Software Engineering Intern (Machine Learning) — *May 2024 – Jul. 2025*
District Hut — *Sacramento, CA*

- Scaled distributed data pipelines processing 3 TB/week across distributed cloud infrastructure, improving throughput and system reliability in a high-volume production environment.
- Optimized data workflows and query performance, identifying and resolving bottlenecks that cut average pipeline latency and improved system stability under sustained load.
- Revamped data integrity checks with SQL, preventing five critical production issues related to data corruption and automating 75% of data validation tasks with 100% test coverage.
- Collaborated with a team of 3 engineers to automate model retraining agents, reducing collective manual effort by 6 hours per week via autonomous performance metrics.

PROJECTS

S.A.I.N.T. — Network Intrusion Detection System | *Python, PyTorch, Flask, Redis, NumPy*

- Partnered with 2 peers to build an AI agent reasoning engine for threat detection, reaching 89% accuracy on NSL-KDD (125K+ samples) through feature engineering and model tuning.
- Designed a decision-making layer using 41 traffic-flow features (protocol stats, TCP flag distributions, temporal patterns) to autonomously classify and flag malicious network activity in real time.
- Architected a low-latency inference agent using Flask + Redis + Plotly, processing 50–100 network connections/sec with an interactive agent-in-the-loop dashboard for analyst review.

StockPulse — Autonomous Stock & Crypto Monitoring Platform | *Python, Flask, Docker, Alpaca API, Binance API, Telegram*

- Built a full-stack autonomous trading monitor integrating 2 live market APIs (Alpaca equities, Binance crypto) with real-time crash detection, ML-based price prediction, and unified portfolio tracking across both asset classes.
- Engineered a multi-agent alert pipeline with Telegram notifications and a job scheduler, enabling 24/7 market surveillance and automated daily reporting with zero manual intervention.
- Containerized the full application with Docker and deployed on cloud infrastructure, serving live model inference with sub-second latency.

PingVault — Network Observability Agent | *Python, ICMP, SNMP, TCP/UDP, PostgreSQL, Alembic*

- Created a production-grade network monitoring tool spanning 3 protocols (ICMP ping, SNMP polling, TCP/UDP port health checks) with agent-based metrics collection and a real-time HTML observability dashboard.
- Designed a modular architecture with Alembic database migrations and structured config management, mirroring enterprise SRE tooling patterns for scalability and maintainability.

TECHNICAL SKILLS

Data & ML: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, model evaluation, SAP (Data Integration)

AI & Agents: Autonomous Agents, LLM Orchestration (LangChain/CrewAI), RAG, Prompt Engineering

Programming: Python, Java, C/C++, SQL, JavaScript, Swift, MATLAB, R

Frameworks/Platforms: React.js, Node.js, Spring Boot, AWS, GCP, Docker, Git

Security/DevOps: SIEM, GitHub Actions, CI/CD, monitoring pipelines